Reg No | | | | | | | | | | |

MADP31

# III SEMESTER ENGINEERING EXAMINATIONS DECEMBER 2015

## Diploma Mathematics I

Time: 3 Hours

Max Marks:70

Instructions: 1. Answer any Five Full Questions

1. a. Find the angle between the radius vector and tangent for the curve $r = a(1+\cos\theta)$ at $\theta = \frac{\pi}{3}$. 04

   b. Find the pedal equation of $r^n = a^n \cos n\theta$. 05

   c. Obtain the Maclaurin's expansion of $\log(1+x)$. 05

2. a. If $u = e^{ax+by} f(ax-by)$, prove that $b\frac{\partial u}{\partial x} + a\frac{\partial u}{\partial y} = 2abu$. 04

   b. If $x = r\cos\theta$ and $y = r\sin\theta$, prove that $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r}\left\{\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2\right\}$ 05

   c. State and prove Euler's theorem on homogeneous function. 05

3. a. Find $\frac{dz}{dt}$ given $z = xy^2 + x^2y$ where $x = at^2$, $y = 2at$. 04

   b. If $u = f(y-z, z-x, x-y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ 05

   c. If $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at (1,-1,0). 05

4. a. Evaluate $\int_{-c}^{c}\int_{-b}^{b}\int_{-a}^{a}(x^2+y^2+z^2)\,dz\,dy\,dx$ 07

   b. Prove that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 07

5. a. Find by double integration the area enclosed by the curve $r = a(1+\cos\theta)$ between $\theta = 0$ and $\theta = \pi$. 07

   b. Show that $\int_a^{\infty} e^{(2ax-x^2)}\,dx = \frac{\sqrt{\pi}}{2}e^{a^2}$ 07

6. a. Find the unit tangent vector to the curve $\vec{r}(t) = 4\sin t\,\hat{i} + 4\cos t\,\hat{j} + 3t\hat{k}$ 04

   b. A particle moves along a curve whose parametric equations are $x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t$. Find the velocity and acceleration at any time t and also their magnitude at t=0. 05

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c. Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2,-1,2)$ along $2\hat{i} - 3\hat{j} + 6\hat{k}$ **05**

7. a. If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div}\vec{F}$ and $\text{curl}\vec{F}$ at the point $(1,-1,1)$. **04**

b. Find the unit normal vector to the surface $x^2y - 2xz + 2y^2z^4 = 10$ at $(2,1,-1)$. **05**

c. Find the value of the constant 'a' such that $\vec{F} = (axy - z^3)\hat{i} + (a-2)x^2\hat{j} + (1-a)xz^2\hat{k}$ is irrotational. **05**

8. a. Find the total work done by the force represented by $\vec{F} = 3xy\hat{i} - y\hat{j} + 2zx\hat{k}$ in moving a particle round the circle $x^2 + y^2 = 4$. **04**

b. Employ Green's theorem in a plane to show that the area enclosed by a plane curve C is $\frac{1}{2}\oint_c xdy - ydx$ and hence find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. **05**

c. Evaluate $\int_s (ax\hat{i} + by\hat{j} + cz\hat{k}).\hat{n}ds$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$. **05**